

columns "A" and "B" are 1 to 5. To create a column "C" which contains the sum of the numbers in columns "A" and "B",
df["C"] = df["A"] + df["B"]
To format df
df2 = df.iloc[[3, 2, 1], [0, 2, 3]]
df2 = df2.iloc[[24, ["A", "C", "D"]], sort_values(df2, "A", ascending = False)
df2 = df2.iloc[[1, 2, 4, ["A", "C", "D"]], sort_values(df2, "C")
class Flight:
def __init__(self, airline):
self.airline = airline
def set_date(self, date):
self.date = date
my_flight = Flight("KLM")
Then I use the combination like
my_flight.set_date(date). #DO NOT include something else such as my_flight.in()
*Also remember that the default values which does not have to be specified necessarily. Date can be non-existent. self there means original book, while other is the second one
To compare the average
def compare(self, other):
avg_better = self.average_score() > other.average_score()
2. for first, second in [(self, other), (other, self)]:
avg_better = first.average_score() > second.average_score()
In the second condition, Here first is self, second is other during the first loop and the other way around for the second loop
* read the condition carefully and see if there is no lacking criterion
To print [NAME] has received a grade of [GRADE]
Students = [{"Name": "Adam", "Grade": 7.5}, {"Name": "Adam", "Grade": 7.5}]
なお内が一のElement (student)として扱われる
for student in students:
print(f'{student["Name"]}
has received a grade of {student["Grade"]:.1f}.')
"Student[Name]" is like indexing ex: x[key] = y to grab "Adam"
if here shows 1 decimal float
To remove "r" from phone numbers (as a string)
def format_phone_number(number):
return number.replace("r", "")
format_phone_number("number")
digits = []
for char in number:
if char in "0123456789":
digits.append(char)
return "".join(digits)
def format_phone_number(number):
return "".join(number.split("-"))
*The point here is that replace does NOT change the original value, so I have to return and reassign in one line
get_tld("https://www.uva.nl/en/education/bachelor-s/bachelors.html")
To return "nl" from the string
url_1 = url_1.split("/") [1] #takes out after www
url_2 = url_1.split("/") [0] #takes out www.uva.nl
url_3 = url_2.split(".") [-1] #takes out nl
2.get_tld(url):
return
url_1.split("/") [1].split("/") [0].split(".") [-1] #do all at once
3.get_tld(url):
url_1 = url[url.find("/") + 2:]
#find() looks up the index of whatever I do, then picks up the starting index, so between : and // . So +2 from there is between // and www and so thatで取れる
url_2 = url_1.split("/") [-1]
return url_2.split(".") [-1]
Anagram question to check if or not
def is_anagram(word1, word2):
return sorted(list(word1.replace(" ", "").lower())) == sorted(list(word2.replace(" ", "").lower()))
This 1.delete white spaces 2.makes all lower cases 3.sorted within a list
.sort() is not for string, but for lists. Also, it changes the original variables, but does not check what is there. Returns None.
.islower() returns T/F while .lower() makes lower cases.
if .replace() is working on a list instead of string, it does not

Do the same thing in case the row labels are from 1 to 5 and the columns are named "A", "D" because iloc refers INDEX while loc refers LABELS directly.
* No need [] in case there is a single letter. However, in case I refer to the column label in a letter, I DO need []
* Check the values for sorting
In case I want only column B and even numbered index
df2 = df2.iloc[df.index % 2 == 0, ["B"]]
Let's say s stands for pandas series. s = "Ams; NH"
"UTR; UTR"
In order to take out only the city names,
1.s.map(lambda x: x.split(",") [0])
2.s.str.split(",") [0]
map here is "for every element"
s.str[9] would work for Amsterdam as it contains 9 characters but not others.
s.str is the same as for string is i.e (for every single pair of city and province). .split(",") creates ["Ams"; "NH"]
"NH"; ... and [0] picks up the city name.
Series is smart enough to calculate directly like
(s - s.mean()) ** 2 to calculate the squared difference between the variable and mean.
When using map, to calculate, always need lambda
s.map(lambda x: (x - s.mean()) ** 2)
(s - s.mean()) .map(lambda x: x**2)
Also show the same, where (s - s.mean()) is treated x
DataFrame df is a variable, so possible to use index as if df is a variable.
df.iloc[[3, 4, 5], 6, "D"] is the same as
df.iloc[[3, 4, 5], 6, "D"] if the values in row 3-6 satisfies the condition
"s" is negation in pandas
For DataFrames, axis 0 is index while axis 1 is other columns. If sort_index(axis=1), in case they are letters or words, it is sorted by alphabetical order.
Datetime
.strftime() creates a datetime object out of a string from datetime import datetime
datetime.strptime(date_str, "%d-%m-%Y")
This shows
.strftime() creates a formatted string out of date, time, datetime objects.
from datetime import datetime
dates.strftime("%d/%m/%Y")
Datetime makes dates possible to calculate
If there is no specification of 単位 like days, the returned value is in milisecond
If I want how many days it has been counting from 1/1
def dat_of_year(dt):
return (dt - datetime(df.year, 1, 1)).days + 1
"dt.year" makes sure to grab the same year
*year, 1 shows Jan 1st
from datetime import datetime, timedelta
date = datetime.strptime("04.05.2020", "%m.%d.%Y")
print(date + timedelta(days=-10).strftime("%d-%m-%Y"))
This returns 26-03-2020.
To compare the dates or times, it is not possible to directly use "12:30" for instance, because they are just strings and not possible to compare
(self.date.year, self.date.month, self.date.day, 12, 30)
It is recognised at 12:30 because of the order.
To create a comprehension, else statement should not come at the end
Mountain's example
mountains = {"Denali": 6190m, "Aconcagua": 6961m, "Kilimanjaro": 5,895m...}
To create
peaks = {"Denali": 6190, "Aconcagua": 6961, "Kilimanjaro": 5895...}
1.peaks = peak.split(",") [0]:
int(peak.split(",") [1]) [-1].replace(" ", "") for peak in mountains
#[-1] removes "m"
2.names = [peak.split(",") [0] for peak in mountains]
heights = [int(peak.split(",") [1].replace(" ", "").replace("m", "")) for peak in mountains]
peaks = dict(zip(names, heights))
= tests T/F so returns 1 or 0
So it is possible to count the number of occurrence
grades = [ID1: grade1, ID2: grade2, ID3: grade3...]
dutch_grades = [1.0, 1.5, ..., 9.5, 10.0]
dutch_curve = sorted(sum(v = grade for v in grades.values())) for grade in dutch grades]
enumerate is for index, even if the name wasn't necessarily so. x.index() shows index as a number.
* If there were multiple numbers which are the same, it might mess up.

To print every even number between 0 and 20
i = 0
while i <= 20:
if i % 2 == 0:
print(i)
i += 1
If the range was (0, 20), it does not include 20
Insert the value which is suggested to check if it gives the value I want.
To make
y = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i']
x = [x[i:i+3] for i in range(0, 7, 3)]
Slicing: if you grab multiple variables at once, it creates a list
In this range, [0, 0+3, 0+3+3, ...]
To pick up the max value of sum within ()
x = [[8, 20, 300], [7, 8, 0], [10, 11, 12], [40, 5, 6]]
To return y = [8, 20, 300]
z = []
for i in x:
z.append(sum(i))
y = x[z.index(max(z))]
*max(x) doesn't work as it is not summed up
*x.sort() doesn't make sense to use here
Scope
If a local variable is used in the global situation, it gives an error
It is a global variable if it is outside of def
Within def, all the variables are considered local
Lambda
x = [1, 2, 3, 4, 5]
To print out elements which is multiplied by 2
[2, 4, 6, 8, 10]
1. list(map(lambda i: i * 2, x))
2. [(lambda i: i * 2)(item) for item in x]
3.y = lambda i: i * 2
(y)(j) for j in x]
map(lambda..., x) is the same as x.map
Dictionary
Check how sorted works in the function.
Write down what is happening.
l1 = [1, 2, 3, 4]
l2 = [2, 3, 4, 5]
d1 = {}
for key, value in zip(l1, l2):
d1[key] = value
d2[key] = value
print(d1 == d2)
Zip function combines two lists as (key, value) as a list, so [(1, 2), (2, 3), (3, 4), (4, 5)] and it is turned into a dict.
For the second for loop, something shown as "key" is actually INDEX. Also, given enumerate(1, 2, 1), the counting of index starts from 1. So the outcome for the both for loops are the same.
for key, value in d1.items():
* In case there is both key and value in a for loop, DO NOT miss items()
list(index + 1) shows the one next
main([1, 3, 2, 4])
To create (1, 1, 3, 3, 2, 2, 4)
Def main(l1):
result = ()
for index, value in zip(l1[-1:], l1[1:]):
result[index] = value
return result
This index is actually just the name. It is simply the first value taken out in zip function.
l1[-1:] takes out 1, 3, 2 while l1[1:] takes out 3, 2, 4.
In case of using while loops, be careful where to state the counter. It has to be placed at the end.
+=, /=, *= are to calculate OVER the result found in the previous loop.
d1 = {1: 2, 2: 1}
result = 1
for x in d1.values():
result *= x
print(result)
1*2 = 2, 2*1 = 2
result = 4
for x in d1.items():
result = result // x[1]
print(result)
4/2 = 2, 2/2 = 1
class Student:
def __init__(self, level):
self.level = level
def __str__(self):
return f"Student {self.level} level {self.level}"
student_1 = Student("bachelor")
student_2 = Student("bachelor")
student_3 = Student("bachelor")
student_4 = Student("bachelor")
student_5 = Student("bachelor")
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